



CODEX

ZHIIHAO YIN

EMBEDDED & IOT ENGINEER

M.Sc. in HCI & B.Eng. in IoT. Expert in C/C++ embedded programming, smart sensing data acquisition, and multi-protocol edge communications, with excellent hands-on hardware/software co-debugging and industrial validation experiences.

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DEGREE: M.Sc. in HCI
GRAD: Dec 2025

HONORS & SKILLS

Competitions & Certifications

- Grand Prize, National Final, 17th Bochuang Cup Embedded Design Competition
- 1st Prize, National Final, Computer Network Technology Challenge
- 3rd Prize, National Final, 15th Chinese College Student Computer Design Competition
- 2nd Prize, Liaoning Computer Design Competition
- CET-6 | IELTS 6.5. Fluent in reading technical datasheets, schematics, and English documentations.

Skill Radar

- Embedded Systems & HW
 - C/C++ Programming
 - MCU Drivers
 - ESP32 / STM32
 - GPIO/UART/SPI
 - I2C/ADC/PWM
 - RTOS Systems
 - Git Control
 - Oscilloscopes
 - Resource Constraints
 - Datasheet Analysis
- Edge-Side Communications & Integration
 - MQTT Protocol
 - Modbus Protocol
 - TCP/IP & WebSockets
 - Edge Intelligence
 - HW/SW Co-Debugging
 - Industrial Field Testing
 - FSM Firmwares
 - Technical Specifications
- Core Attributes & Teamwork
 - EHS Hazard Inspections
 - Cross-Functional Collab.
 - Abstraction & Logic
 - Milestone Reviews
 - Standard Work Instructions
 - English Datasheet Reading

EDUCATION

2024.09 - 2025.09 UNIVERSITY OF NOTTINGHAM (UK)

HCI

M.Sc. in Human-Computer Interaction | Master's Degree

Core Courses: Smart Spatial Interaction, Mixed Reality (MR) Integration, Edge Intelligence Computing, Advanced HW/SW Co-Debugging, Usability Evaluation under Resource Constraints, Human Performance Research.

- Academic Project - VR Rhythm Game Spatial Prototype: Headed development of an audio-reactive VR spatial interaction prototype; implemented C++ multi-threaded data filtering and low-latency serial UART communications to achieve real-time sensor data fusion under strict MCU memory constraints, isolating latency bottlenecks.
- Academic Project - Sensor Data Modeling & Technical Spec Writing: Led capture of high-frequency physical motion sensors. Formulated interaction paradigms, drafted technical specifications, and coordinated milestone reviews to close hardware loop issues.

IOT

2020.09 - 2024.07 DALIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Eng. in Internet of Things Engineering | Bachelor's Degree

Ranked 1st | GPA 4.1/5.0

Academic Honors: First-Class Golden University Scholarship recipient.
Core Courses: Embedded System Design, MCU Principles & Peripherals (GPIO/UART/SPI/I2C/ADC), RTOS (Real-Time Operating Systems), C/C++ Programming, Wireless Sensor Networks (WSN).

- Academic Project - IoT Industrial Production Line Management System: Independently designed and implemented an industrial production data node on ESP32 microcontrollers. Programmed C/C++ drivers for peripheral interfaces, managed multi-thread scheduling in RTOS, and established low-latency MQTT channels, logging all debugging logs and change histories.

WORK EXPERIENCE

2026.02 - 2026.05 MIAO INC. (SHANGHAI)

EMBEDDED / IOT

Embedded Systems & IoT R&D Engineer

Core Responsibilities: Directed edge co-debugging pipelines, C++ state machine firmwares, and multi-protocol wired/wireless sensing data acquisitions for multiplayer titles "Five-Star Citizen" and "Bombing Blind Shot".

- Edge-Side Data Acquisition & Multi-Protocol Debugging: Collaborated with hardware and QA teams to co-debug sensing prototypes. Configured SPI, I2C, and wired/wireless communications (UART serial, WebSockets, TCP/UDP) to achieve real-time, low-latency filtering, buffering, and packet transmission of high-frequency physical inputs; utilized oscilloscopes and logic analyzers for time-domain physical signal and timing audits to isolate latency bottlenecks and timing noise, filing standard changelogs and debugging records.
- "Bombing Blind Shot" HW/SW Co-Debugging & Edge Intelligence: Maintained edge node processing pipelines and custom MCP server communication middlewares connecting local LLM agents to Roblox runtimes. Directed integration tests and pilot validation, securing high reliability under peak server loads, while strictly adhering to laboratory EHS safety guidelines and ESD static protections to mitigate electrical risks during live-circuit debugging.
- Git Version Management & Technical Review Standards: Supervised branch deployments and code reviews using Git. Programmed automated regression scripts to guarantee microcontroller firmware compatibility, establishing standard defect tracking procedures to ensure fast resolution and milestone closures.

SERVER DEV

2023.09 - 2024.02

SHANGHAI HUICHENG TECHNOLOGY CO., LTD. (DIGITAL TWIN R&D CENTER)

Embedded & Server R&D Intern

- Edge Gateway Design & Modbus/MQTT Integration: Programmed STM32 and ESP32 microcontrollers, configuring I2C, SPI, RS485, and ADC peripheral interface drivers for real-time high-frequency telemetry data acquisition and digital filtering of industrial sensors (vibration, temperature); developed and debugged industrial telemetry protocols (Modbus-RTU/TCP, MQTT) and peripheral drivers, supporting onsite deployment, multi-node device-level cascade testing, and field communication quality isolations.
- Code Quality Standards & Version Controls: Strictly adhered to software engineering and strict information security guidelines; maintained platform branches, analyzed concurrency issues to optimize telemetry queue buffers, proactively reported firmware defects, and authored 10+ detailed technical specs, API documentations, and code reviews to close system bugs.

PROJECT EXPERIENCE

2025.01 - 2025.06 WARHAMMER DIGITAL LINE USER RESEARCH

USER RESEARCH

System Usability & Co-Debugging Analyst (University of Nottingham Campus Collaboration Project)

Responsibilities: Led player research for the Games Workshop Warhammer digital ecosystem. Utilized mixed-method studies to diagnose and optimize interactive UI flow pain points.

- Multi-Platform Communication & Interface Auditing: Supervised API network performance, protocol latency, and transmission usability audits between mobile client interfaces and backend servers. Utilized packet analyzers to diagnose telemetry jitter and packet drops, isolating interface software bugs to compile comprehensive technical proposals and API specifications.
- Cross-Functional Review & UGUI Optimization: Led milestone reviews and release change control workflows with Games Workshop's digital division, pushing interface optimization specs to restructure progressive onboarding flows, successfully closing high-risk defects and securing high system stability.

2025.03 - 2025.06

"MILLENNIUM MIST" - GAMEPLAY VALIDATION PROTOTYPE

UNITY / C#

Solo Unity & C++ Engine Prototype Developer

Responsibilities: Built an action-shmup prototype from scratch in C++/C# to validate high-performance business logics and combat frameworks under resource constraints.



C++ Business Logic & HFSM Firmware Design: Independently programmed and validated prototype control logic under highly constrained system resources (CPU/RAM/Flash limits); engineered Hierarchical Finite State Machines (HFSM) to govern complex combat rules, utilizing C++ multi-threaded optimizations and custom memory pools to eliminate memory fragmentation and secure sub-millisecond response rates.



Tactile Actuator Calibration & Performance Benchmarking: Configured and debugged physical haptic feedback actuators via UART, GPIO interrupts, and PWM drivers; utilized logic analyzers for strict time-domain verification of haptic feedback loops and sensor inputs, ensuring microsecond response rates even under peak computational loads to close HW/SW integration loops.

XR / HCI

2025.07 - 2025.10

"MR RECIPES" - MIXED REALITY SPATIAL INTERACTION PROTOTYPE

Meta Quest 3 Smart Sensing & Spatial Positioning R&D

Responsibilities: Built a spatial cooking assistant for Meta Quest 3 combining sensor inputs and real-time physical overlays.



Sensor Calibration & Spatial Anchor Alignments: Deployed multi-sensor high-frequency telemetry data fusion (high-precision IMU and optical hand-tracking sensors) on Meta Quest 3; designed custom Kalman and sliding window noise-reduction filters to mitigate spatial infrared noise, securing sub-millisecond low-latency virtual-physical spatial alignments in real kitchen setups.



Pilot Onsite Testing & Defect Loop Closures: Conducted contextual inquiries and high-fidelity pilots in physical environments; adhered to onsite EHS safety protocols and operational guidelines, isolated and resolved sensor accuracy failures (due to high temperature and moisture) and radio attenuation via firmware fault-tolerance optimization, closing integration defects with the hardware team.

ACTIVITIES

STAGE SUPPORT

Long-term Active

LOGISTICS & STAGE SUPPORT VOLUNTEER

Prominent Music Festivals (Midi, Aranya, Bubble Land, etc.)

Responsibilities: Worked backstage under rigorous EHS guidelines to enforce safety standards, evaluate electricity risks, and maintain zero-accident operations.



Safety Audits & Lightning Proofing Suggestions: Actively inspected backstage power cabinets and wiring layouts; proposed 10+ lightning and electrical leak prevention protocols to ensure absolute event security.

MAKER SPACE

2022.06 - 2023.04

PRESIDENT OF MAKER SPACE

University Interdisciplinary Technology Association



Embedded Lab Instruction & Lab Standardizations: Supervised hardware selections, I/O pin layouts, and serial UART communication firmwares on ESP32/STM32 platforms for 100+ members. Secured sponsorships to enforce laboratory safety guidelines.